

BLIP and Captioning

DATA 443/643 · Week 5, note 5.2

Week 5

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Ideas from Stanford CS231N 2025 L16 (CoCa: add a decoder) and the BLIP paper (ITC, ITM, LM, bootstrap). Original slides; those sources are not copied.

Lecture plan

- 1 Retrieve versus write
- 2 Filter noisy web pairs
- 3 Three losses: ITC, ITM, LM
- 4 One ITM numeric

Key goal: CLIP scores; BLIP can decode; the filter is why web alt-text is not the training text.

First time: what / why / formula

- What / why. Three losses: ITC (contrastive), ITM (matched vs unmatched pair), LM (caption tokens). Bootstrap: generate captions, filter, train. CLIP cannot decode a sentence. Naive captioners train on noisy alt-text. BLIP tries to clean the web and add generation.
- Architecture. Image encoder, text encoder, and a decoder with cross-attention into image tokens (queries from text, keys/values from vision).
- Formula. ITC as InfoNCE; ITM as logistic on a CLS; LM as $\sum_t -\log p(w_t \mid w_{<t}, \text{image})$.
- Tradeoffs. + Captions and retrieval in one family. – Hallucinated captions; filter is extra machinery; heavier than frozen CLIP.

How. ITC aligns. ITM is a binary match head. LM is next-token on the caption. Generation uses the decoder; retrieval can use ITC embeddings.

Two jobs that look similar in a demo

Retrieval (CLIP)

Return a string that already lives in the gallery. Metric: recall@k.

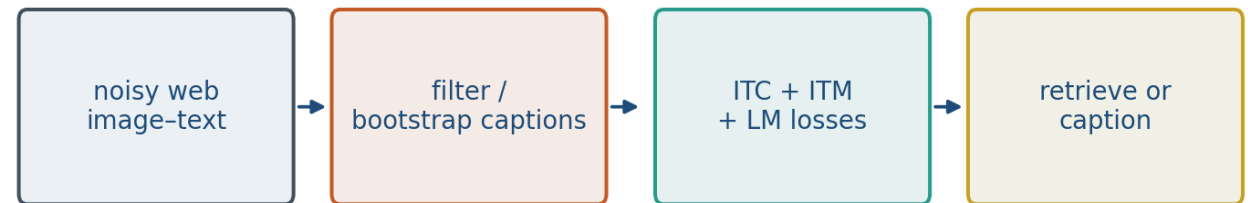
Captioning (BLIP)

Generate a new string. Metric: CIDEr, CLIP-score, and a human check. Hallucinated objects are the failure.

Bootstrap, then train

- Alt-text is often a filename or a SKU.
- Generate a caption, keep it if ITM still matches.
- CoCa adds a decoder. BLIP also cleans the pairs.

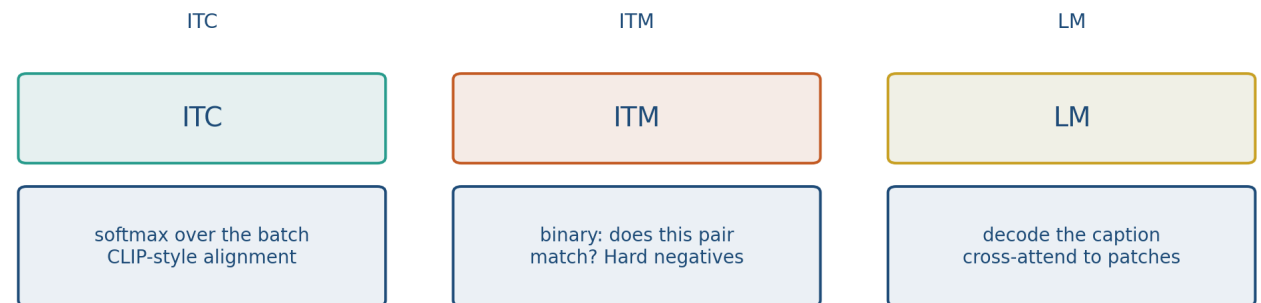
BLIP: clean the pairs, then learn to match and to write



Three losses, one backbone

- ITC: CLIP-style softmax over the batch.
- ITM: yes/no on one pair, often a hard negative.
- LM: next-token, cross-attend to ViT patches.

BLIP keeps three losses. CoCa (CS231N) adds a decoder; BLIP also filters the web.



ITM is a sigmoid, not a batch softmax

$$\sigma(s) = \frac{1}{1 + e^{-s}}$$

Matched $S = 2$: $\sigma \approx 0.88$,
CE ≈ 0.13 .

Mismatch $S = -1$:
 $\sigma \approx 0.27$, CE ≈ 0.31 .

ITC needs a batch. ITM can score one pair.

Write the ITM table

ITM is not a batch softmax. One pair, a yes/no score.

- True caption: a red mug on a desk.
- False caption: a blue bicycle.
- If you only retrieve, stop at ITC. If you write, you need LM.

ITM logit	sigmoid	binary CE term
matched $s = 2$	$\sigma(2) \approx 0.88$	$-\log \sigma \approx 0.13$
mismatch $s = -1$	$\sigma(-1) \approx 0.27$	$-\log(1 - \sigma) \approx 0.31$

What to take from this lecture

Filter

Do not train LM on DSC0001.jpg.

Three verbs

Align, match, decode. Different losses.

Whisper clip

Encoder–decoder analog. Whisper has no ITC+ITM.